

Bruno Gehlen Ferreira da Silva

Full-Stack & AI Developer

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PROFESSIONAL SUMMARY

Full-stack developer with a strong analytical foundation (B.Sc. in Physics, M.Sc. candidate in Mathematics at USP) and professional experience building data-intensive systems for the financial sector. Helped engineered ETL pipelines, reconciliation engines, and automation tooling for a core banking platform at Santander (via Management Solutions) and and currently develop AI agent-powered automated platforms for various companies in the market. Combines production-grade back-end and data engineering skills (PySpark, SQL, SciPy) with modern front-end expertise (React, Next.js, Tailwind) and hands-on cloud/DevOps & CI/CD practice (Git, Docker, AWS, self-hosted infrastructure). Focused on applying this full-stack toolkit to AI-powered products, with working knowledge of machine learning libraries and LLM tooling.

WORK EXPERIENCE

Software & Data Engineering Consultant (A2) - Management Solutions Aug 2025 - Mar 2026

Client: Santander - Core Banking System Project · São Paulo, SP

- Colaborated to engineer and deploy robust ETL pipelines automating ingestion, transformation, and validation of financial data from capital markets platforms into a centralized Operational Data Store (ODS).
- Led data migration and integration of a core trading platform and implemented adaptations for new financial products, ensuring accurate data flow from front-office trade capture to back-office accounting ledgers.
- Co-authored technical specifications and validation strategies for core data services and reconciliation engines, translating complex accounting requirements from business stakeholders into scalable technical solutions.

TECHNICAL SKILLS

Front-end	React, Tailwind CSS, Figma, WCAG accessibility
Back-end	Node.js, Python, FastAPI, REST APIs, PostgreSQL
Data & AI	PySpark, SciPy, Scikit-learn, LangChain
DevOps & Cloud	Docker, Git, NGINX, AWS, Google Cloud, Azure, Supabase

PROJECTS

Cade o Circular Hiatus

- Developed a monitoring web app for USP bus lines using the SPTrans API, combining frontend development, API integration, and data-driven UI work.

Hackerspace IFUSP Ongoing

- Developing the official website for the Hackerspace IFUSP community, combining modern frontend implementation, content structure, and a polished user experience for public outreach.

Gnosis Ongoing

- Developing a universal AI GraphRAG memory system for LLMs, combining vector and graph databases to create real "memory" for LLMs, with a focus on reasoning and long-term knowledge retention.

EDUCATION

M.Sc. in Mathematics - IME-USP, University of São Paulo Apr 2025 - Present

Research in geometric analysis and nonlinear PDEs focusing on singular solutions to the conformal Riemannian metrics Yamabe problem, with applications to mathematical physics.

B.Sc. in Physics - IF-USP, University of São Paulo Apr 2021 - Dec 2024

Undergraduate research in quantum field theory (IFT-UNESP) and statistical/thermal models of heavy-ion collisions (IF-USP), including scientific computing in Wolfram Mathematica and Python.